

## Integrals involving Fourier basis functions

The below theorem is used everywhere.

*Theorem 0.1 (Fourier Integral):*

$$\int_0^{2\pi} e^{imx} dx = \begin{cases} 2\pi(m=0) \\ 0 \quad (m \in \mathbb{Z} \setminus \{0\}) \end{cases} \quad (1)$$

*Proof:*

$$\int_0^{2\pi} e^{imx} dx = \frac{e^{im \cdot 2\pi} - e^{im \cdot 0}}{im} \quad (2.1)$$

$$= \begin{cases} 2\pi(m=0) \\ 0 \quad (m \in \mathbb{Z} \setminus \{0\}) \end{cases} \quad (2.2)$$

□

- Integral equations coming from common partial differential equations in  $\mathbb{S}^1$  ( $\mathbb{R}^2$ ) often involve singular integrals, derivative of integrals.
- To compute these integrals, exact values of integrals of Fourier basis functions are needed.
- To deal with the singularity while maintaining  $2\pi$ -periodicity, we need a function which only has singularity at  $0(2\pi)$  and is  $2\pi$ -periodic.
  - $\csc^n t, \cot^n t, n \in \mathbb{N}$ : singularities at  $\pi$  as well
  - $\csc^n \frac{t}{2}, n \notin 2\mathbb{N}$ : not  $2\pi$ -periodic
  - $\cot^n \frac{t}{2}, n \in \mathbb{N}$ : ok
  - $\csc^{2n} \frac{t}{2}, n \in 2\mathbb{N}$ : ok

*Definition 0.1 (Cauchy Principle value on a circle):*  $\forall \alpha > 0. \forall t \in [0, 2\pi). \forall f \in C^{0,\alpha}(0, 2\pi)$

$$\oint_0^{2\pi} f(x) \cot\left(\frac{x-t}{2}\right) dx := \lim_{\varepsilon \rightarrow 0} \left( \int_{\varepsilon}^{2\pi-\varepsilon} f(x+t) \cot\left(\frac{x}{2}\right) dx \right) \quad (3)$$

*Definition 0.2 (Hadamard Finite-Part on a circle):*  $\forall \alpha > 0. \forall t \in [0, 2\pi). \forall n \in \mathbb{N} \setminus \{1\}. \forall f \in C^{n-1,\alpha}(0, 2\pi).$

$$\oint_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) dx := \lim_{\varepsilon \rightarrow 0} F_0(\varepsilon) \quad (4)$$

where

$$\forall \varepsilon \in (0, \pi). \int_{\varepsilon}^{2\pi-\varepsilon} f(x+t) \cot^n\left(\frac{x}{2}\right) dx = F_0(\varepsilon) + F_1(\varepsilon) \quad (5.1)$$

$$\lim_{\varepsilon \rightarrow 0} F_0(\varepsilon) < \infty \quad (5.2)$$

$$\lim_{\varepsilon \rightarrow 0} \log\left(4 \sin^2\left(\frac{\varepsilon}{2}\right)\right) \frac{1}{F_1(\varepsilon)} < \infty \quad (5.3)$$

*Theorem 0.2 (Hadamard Finite-Part on a circle):*  $\forall n \in \mathbb{N}. \forall f \in C^{n,\alpha}(\mathbb{R}/2\pi) \cap L^{1+}(\mathbb{R}/2\pi).$  Let

$$(T_n f)(t) := \oint_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) dx \quad (6)$$

then

$$(T_{n+1}f)(t) := -\frac{2}{n} \frac{d}{dt}(T_n f)(t) - (T_{n-1}f)(t) \quad (7)$$

where

$$(T_1 f)(t) := \oint_0^{2\pi} f(x) \cot\left(\frac{x-t}{2}\right) dx \quad (8.1)$$

$$(T_0 f)(t) := \int_0^{2\pi} f(x) dx \quad (8.2)$$

*Theorem 0.3:*  $\forall n \in \mathbb{N}. \forall f \in C^{n,\alpha}(\mathbb{R}/2\pi) \cap L^{1+}(\mathbb{R}/2\pi)$ .

$$\frac{d}{dt} \oint_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) dx = \oint_0^{2\pi} f'(x) \cot^n\left(\frac{x-t}{2}\right) dx \quad (9)$$

*Proof:*

$$\frac{d}{dt} \oint_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) dx = \frac{d}{dt} \oint_0^{2\pi} f(x+t) \cot^n\left(\frac{x}{2}\right) dx \quad (10.1)$$

$$= \frac{d}{dt} \lim_{\varepsilon \rightarrow 0} F(t, \varepsilon) \quad (10.2)$$

$$\stackrel{?}{=} \lim_{\varepsilon \rightarrow 0} \frac{d}{dt} F(t, \varepsilon) \quad (10.3)$$

$$= \oint_0^{2\pi} f'(x) \cot^n\left(\frac{x-t}{2}\right) dx \quad (10.4)$$

□

*Theorem 0.4:*

$$\oint_0^{2\pi} f'(t)g(t) dt \stackrel{?}{=} - \oint_0^{2\pi} f(t)g'(t) dt \quad (11)$$

*Theorem 0.5:* Let

$$I_{m,n} = \oint_0^{2\pi} e^{imt} \cot^n\left(\frac{t}{2}\right) dt \quad (12)$$

then following recurrence relation holds:

$$\forall m \in \mathbb{Z}. \forall n \in \mathbb{N} \setminus \{1\}. I_{m,n} = \frac{2im}{n-1} I_{m,n-1} - I_{m,n-2} \quad (13)$$

with initial values:

$$I_{m,0} = 0, \quad I_{m,1} = 2\pi i \operatorname{sgn}(m) \quad (m \neq 0) \quad (14.1)$$

$$I_{0,0} = 2\pi, \quad I_{0,1} = 0 \quad (14.2)$$

*Proof:*

- $I_{m,0}, m \in \mathbb{Z}$ : Theorem 0.1.
- $I_{0,1} = 0$ : Due to the asymmetry of integrand  $\cot \frac{t}{2}$ .
- $I_{m,1}, m \in \mathbb{Z}$ : Follows [1, Lemma 8.23.].
  - $\cot \frac{t}{2} = \frac{\cos \frac{t}{2}}{\sin \frac{t}{2}} = i \frac{e^{\frac{it}{2}} + e^{-\frac{it}{2}}}{e^{\frac{it}{2}} - e^{-\frac{it}{2}}} = i \frac{e^{it} + 1}{e^{it} - 1}$

- $\therefore \forall m \in \mathbb{N}. (e^{imt} - 1) \cot\left(\frac{t}{2}\right) = (e^{it} - 1) \left( \sum_{j=0}^{m-1} e^{ijt} \right) \cot\left(\frac{t}{2}\right) = i(e^{it} + 1) \sum_{j=0}^{m-1} e^{ijt}.$
- $\therefore \forall m \in \mathbb{N}. I_{m,1} - I_{0,1} = 2\pi i$  Integrating both hands and Theorem 0.1
- $\therefore \forall m \in \mathbb{N}. I_{m,1} = 2\pi i$
- $\forall m \in \mathbb{N}, I_{-m,1} = \overline{I_{m,1}} = -2\pi i$
- Recurrence relation: Due to Theorem 0.2 Theorem 0.3 and  $(e^{imt})' = ime^{imt}.$

□

*Theorem 0.6:* Let

$$J_{m,n} = \oint_0^{2\pi} e^{imt} \log\left(4 \sin^2\left(\frac{t}{2}\right)\right) \cot^n\left(\frac{t}{2}\right) dt \quad (15)$$

then following recurrence relation holds:

$$\forall m \in \mathbb{Z}. \forall n \in \mathbb{N} \setminus \{1\}. J_{m,n} = \frac{2im}{n-1} J_{m,n-1} - J_{m,n-2} + \frac{2}{n-1} I_{m,n} \quad (16)$$

with initial values:

$$J_{m,0} = -\frac{2\pi}{|m|}, \quad J_{m,1} = 2\pi i \operatorname{sgn}(m) \left( 2H_{|m|} - \frac{1}{|m|} \right), \quad H_m := \sum_{k=1}^m \frac{1}{k} \quad (m \neq 0) \quad (17.1)$$

$$J_{0,0} = 0, \quad J_{0,1} = 0 \quad (17.2)$$

*Proof:*

- $J_{m,0}, m \in \mathbb{Z}$ : Follows [1, Lemma 8.23.].
- $J_{0,1}$ : Due to the asymmetry of integrand  $\log(4 \sin^2(\frac{t}{2})) \cot(\frac{t}{2})$ .
- $J_{m,1}, m \in \mathbb{Z} \setminus \{0\}$ :
  - $L(t) := \log(4 \sin^2(\frac{t}{2})) = -\sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{e^{ikt}}{|k|}$
  - $\forall m \in \mathbb{N}. J_{m,1} = \oint_0^{2\pi} e^{imt} L(t) L'(t) dt = -\frac{im}{2} \int_0^{2\pi} e^{imt} L^2(t) dt =$   
 $-\frac{im}{2} \int_0^{2\pi} e^{imt} \left( \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{e^{ikt}}{|k|} \right) \left( \sum_{l \in \mathbb{Z} \setminus \{0\}} \frac{e^{ilt}}{|l|} \right) dt = (2\pi) \sum_{k,l \in \mathbb{Z} \setminus \{0\}, k+l+m=0} \frac{1}{|k||l|} =$   
 $(2\pi) \sum_{k \in \mathbb{Z} \setminus \{0\}, l \in \mathbb{Z}, k+l+m=0} \frac{1}{|k||l|} = (2\pi) \sum_{k \in \mathbb{Z} \setminus \{0, -m\}} \frac{1}{|k||m+k|}$
  - $\forall m \in \mathbb{N}. \sum_{k \in \mathbb{Z} \setminus \{0, -m\}} \frac{1}{|k||m+k|} = \sum_{k=1}^{\infty} \frac{1}{k(m+k)} + \sum_{k=-m+1}^{-\infty} \frac{1}{|k||m+k|} +$   
 $\sum_{k=-\infty}^{-m+1} \frac{1}{|k||m+k|} = \sum_{k=1}^{\infty} \frac{1}{k(m+k)} + \sum_{k=1}^{m-1} \frac{1}{k(m-k)} + \sum_{k=1}^{\infty} \frac{1}{k(m+k)} = \frac{2}{m} \sum_{k=1}^{\infty} \left( \frac{1}{k} - \frac{1}{m+k} \right) +$   
 $\frac{1}{m} \sum_{k=1}^{m-1} \left( \frac{1}{k} + \frac{1}{m-k} \right) = \frac{2}{m} \left( \sum_{k=1}^m \frac{1}{k} + \sum_{k=1}^{m-1} \frac{1}{k} \right) = \frac{2}{m} (2H_m - \frac{1}{m})$

□

*Theorem 0.7 (Estimates for  $I_{m,n}, J_{m,n}$ ):*

1.  $\forall m \in \mathbb{Z} \setminus \{0\}. \forall n \in \mathbb{N}_0. |I_{m,n}| \leq 2\pi n |m|^{n-1}$
2.  $\forall m \in \mathbb{Z} \setminus \{0\}. \forall n \in \mathbb{N}_0. |J_{m,n}| \leq 2\pi(n+1)^2 |m|^n$

*Proof:* Proved by induction for  $n$

1.  $I_{m,n}$ 
  - $n = 0$ :  $|I_{m,0}| = 0$
  - $n = 1$ :  $|I_{m,1}| = 2\pi$
  - $n > 1$ :  $|I_{m,n}| = \left| \frac{2im}{n-1} I_{m,n-1} - I_{m,n-2} \right| \leq \frac{2|m|}{n-1} |I_{m,n-1}| + |I_{m,n-2}| \leq \frac{2|m|}{n-1} \cdot 2\pi(n-1) |m|^{n-2} + 2\pi(n-2) |m|^{n-3} \leq 4\pi |m|^{n-1} + 2\pi(n-2) |m|^{n-3} \leq 2\pi n |m|^{n-1}$
2.  $J_{m,n}$ 
  - $n = 0$ :  $|J_{m,0}| = \frac{2\pi}{|m|} \leq 2\pi$
  - $n = 1$ :  $|J_{m,1}| = 2\pi \left| 2H_{|m|} - \frac{1}{|m|} \right| \leq 8\pi |m|$

$$\begin{aligned} \bullet \quad n > 1: |J_{m,n}| &= \left| \frac{2im}{n-1} J_{m,n-1} - J_{m,n-2} + \frac{2}{n-1} I_{m,n} \right| \leq \frac{2|m|}{n-1} |J_{m,n-1}| + |J_{m,n-2}| + \\ &\quad \frac{2}{n-1} |I_{m,n}| \leq \frac{2|m|}{n-1} \cdot 4\pi n^2 |m|^{n-1} + 4\pi(n-1)^2 |m|^{n-2} + 4\pi \frac{n}{n-1} |m|^{n-1} \leq 4\pi(4n + \\ &\quad (n-1)^2 + 2) |m|^n = 4\pi(n^2 + 2n + 1) |m|^n = 4\pi(n+1)^2 |m|^n \end{aligned}$$

□

### Subspace $U_N$

*Lemma 0.1:*  $\forall N' \in \mathbb{N}. t_j := \frac{2\pi j}{N'}. \forall m \in \mathbb{Z}. \forall t_{\text{start}} \in [0, 2\pi).$

$$\frac{2\pi}{N'} \sum_{j=0}^{N'-1} e^{im(t_j - t_{\text{start}})} = \begin{cases} 2\pi(m \equiv 0 \pmod{N'}) \\ 0 \quad (\text{otherwise}) \end{cases} \quad (18)$$

*Definition 0.3:*  $U_N := \text{span}(\{e^{imx} \mid m \in \mathbb{Z}, |m| < N\})$

*Theorem 0.8 (Trapezoidal Rule for  $U_N$ ):*  $\forall N \in \mathbb{N}. N$ -point trapezoidal rule is exact for  $U_N$ .

*Proof:* Lemma 0.1 □

*Lemma 0.2 (Trapezoidal Rule for inner product for  $U_N$ ):* Let  $N' := \dim U_N = 2N - 1$ . Let  $t_j := \frac{2\pi j}{N'}$  for  $j = 0, \dots, N' - 1$ .  $\forall f, g \in U_N$ .

$$\int_0^{2\pi} f(t)g(t) dt = \frac{2\pi}{N'} \sum_{j=0}^{N'-1} f(t_j)g(t_j) \quad (19)$$

*Proof:* Since,  $f(t)g(t) \in U_{2N-1}$  Theorem 0.8 can be applied with  $N' = 2N - 1$  points. □

### Quadratures

Now we consider the way to express discrete Fourier expansion of a function in  $U_N$  using the values of the function at  $t_{\text{start}}, t_{\text{start}} + \frac{2\pi}{N'}, t_{\text{start}} + \frac{4\pi}{N'}, \dots$

*Lemma 0.3:*  $\forall N \in \mathbb{N}. \forall f \in U_N. N := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi).$

$$f(x) = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) \cdot \left( \frac{1}{N'} \sum_{|m| < N} e^{-im(t_j + t_{\text{start}})} e^{imx} \right) \quad (20)$$

*Proof:*

$$f(x) = \sum_{|m| < N} \int_0^{2\pi} f(t) \frac{e^{-imt}}{\sqrt{2\pi}} dt \cdot \frac{e^{imx}}{\sqrt{2\pi}} \quad (21.1)$$

$$\stackrel{\text{Lemma 0.2}}{=} \frac{1}{2\pi} \sum_{|m| < N} \frac{2\pi}{N'} \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) e^{-im(t_j + t_{\text{start}})} \cdot e^{imx} \quad (21.2)$$

$$= \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) \cdot \left( \frac{1}{N'} \sum_{|m| < N} e^{-im(t_j + t_{\text{start}})} e^{imx} \right) \quad (21.3)$$

□

*Definition 0.4 (Lagrange basis):*

$$L_j^{t_{\text{start}}}(x) := \frac{1}{N'} \sum_{|m| < N} e^{-im(t_j + t_{\text{start}})} e^{imx} \quad (22)$$

*Theorem 0.9:*

$$\int_0^{2\pi} L_i^{t_{\text{start}}}(t) \overline{L_j^{t'_{\text{start}}}(t)} dt = \quad (23)$$

*Proof:*

$$\int_0^{2\pi} L_i^{t_{\text{start}}}(t) \overline{L_j^{t'_{\text{start}}}(t)} dt = \int_0^{2\pi} \left( \frac{1}{N'} \sum_{|m|<N} e^{-im(t_i+t_{\text{start}})} e^{imt} \right) \left( \frac{1}{N'} \sum_{|n|<N} e^{in(t_j+t'_{\text{start}})} e^{-in(t)} \right) dt \quad (24.1)$$

$$= \frac{1}{N'^2} \sum_{|m|<N} \sum_{|n|<N} e^{-im(t_i+t_{\text{start}})} e^{in(t_j+t'_{\text{start}})} \int_0^{2\pi} e^{i(m-n)t} dt \quad (24.2)$$

$$= \frac{1}{N'^2} \sum_{|m|<N} \sum_{|n|<N} e^{-im(t_i+t_{\text{start}})} e^{in(t_j+t'_{\text{start}})} \delta_{m,n} 2\pi \quad (24.3)$$

$$= \frac{2\pi}{N'^2} \sum_{|m|<N} e^{im(t_j+t'_{\text{start}}-t_i-t_{\text{start}})} \quad (24.4)$$

□

Combining the above lemma with the exact integral values of Fourier basis functions, we can exactly compute the integrals of functions in  $U_N$  multiplied by  $\cot^n(\frac{t}{2})$  or  $\log(4 \sin^2(\frac{t}{2})) \cot^n(\frac{t}{2})$ , by using only the values of the function at  $t_{\text{start}}, t_{\text{start}} + \frac{2\pi}{N'}, t_{\text{start}} + \frac{4\pi}{N'}, \dots$

*Theorem 0.10 (Generalized Garrick–Wittich quadrature for  $U_N$ ):*  $\forall n \in \mathbb{N}_0. \forall N \in \mathbb{N}. \forall f \in U_N. N' := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi).$

$$\oint_0^{2\pi} f(t) \cot^n\left(\frac{t}{2}\right) dt = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) P_j^{N',n} \quad (25)$$

$$P_j^{N',n,t_{\text{start}}} := \frac{1}{N'} \sum_{|m|<N} I_{m,n} e^{-im(t_j+t_{\text{start}})} \quad (26)$$

*Theorem 0.11 (Generalized Kussmaul–Martensen (Kress) quadrature for  $U_N$ ):*  $\forall n \in \mathbb{N}_0. \forall N \in \mathbb{N}. \forall f \in U_N. N' := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi).$

$$\oint_0^{2\pi} f(t) \log\left(4 \sin^2\left(\frac{t}{2}\right)\right) \cot^n\left(\frac{t}{2}\right) dt = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) Q_j^{N',n} \quad (27)$$

$$Q_j^{N',n,t_{\text{start}}} := \frac{1}{N'} \sum_{|m|<N} J_{m,n} e^{-im(t_j+t_{\text{start}})} \quad (28)$$

## Error estimates

## Integral Equations

*Theorem 0.12:*  $X$ : Banach space,  $\forall A \in B(X). \forall T \in B^{\text{inv}}(X). \|T^{-1}(T - A)\| < 1 \implies A \in B^{\text{inv}}(X)$

*Proof:* Neumann series implies  $(I - (T^{-1}(A - T))) \in B^{\text{inv}}(X)$  and

$$\|A^{-1}\| = \|(I - (T^{-1}(A - T)))^{-1}\| \leq \frac{1}{1 - \|T^{-1}(T - A)\|} \quad (29)$$

□

*Definition 0.5:*

$$I_N : \rightarrow U_N, f \mapsto \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) L_{t_{\text{start}}}(x) \quad (30)$$

*Theorem 0.13:* Let  $\varphi$  solution of

$$\varphi(s) + \sum_{n=0}^M \int_0^{2\pi} K_n(s, t) \cot^n\left(\frac{s-t}{2}\right) \varphi(t) dt + \sum_{n=0}^M \int_0^{2\pi} L_n(s, t) \log\left(4 \sin^2\left(\frac{s-t}{2}\right)\right) \cot^n\left(\frac{s-t}{2}\right) \varphi(t) dt = f(s)$$

and  $\varphi' \in U_N$  solution of

$$\varphi(s) + I_N \sum_{n=0}^M \int_0^{2\pi} K_n(s, t) \cot^n\left(\frac{s-t}{2}\right) \varphi(t) dt + I_N \sum_{n=0}^M \int_0^{2\pi} L_n(s, t) \log\left(4 \sin^2\left(\frac{s-t}{2}\right)\right) \cot^n\left(\frac{s-t}{2}\right) \varphi(t) dt = I_N f(s)$$

assume that Equation 31 is solvable. Then

1.  $\exists N_s \in \mathbb{N}. \|T - A\| < \|T^{-1}\|^{-1}$
2.  $\forall N_s [\text{satisfies above condition}]. \forall N \geq N_s. \text{Equation 32 has a unique solution}$

*Theorem 0.14:*

$$\varphi(t_j + t_{\text{start}}) + \sum_{n=0}^M \sum_{k=0}^{N'-1} K_n(t_j + t_{\text{start}}, t_k + t'_{\text{start}}) P_j^{N', n, t'_{\text{start}}} \varphi(t_k + t'_{\text{start}}) \quad (33.1)$$

$$+ \sum_{n=0}^M \sum_{k=0}^{N'-1} L_n(t_j + t_{\text{start}}, t_k + t'_{\text{start}}) Q_j^{N', n, t'_{\text{start}}} \varphi(t_k + t'_{\text{start}}) = f(t_j + t_{\text{start}}) \quad (33.2)$$

*Proof:*

$$\varphi(t_j + t_{\text{start}}) + \sum_{n=0}^M \sum_{k=0}^{N'-1} K_n(t_j + t_{\text{start}}, t_k + t'_{\text{start}}) P_j^{N', n, t'_{\text{start}}} \varphi(t_k + t'_{\text{start}}) \quad (34.1)$$

$$+ \sum_{n=0}^M \sum_{k=0}^{N'-1} L_n(t_j + t_{\text{start}}, t_k + t'_{\text{start}}) Q_j^{N', n, t'_{\text{start}}} \varphi(t_k + t'_{\text{start}}) = f(t_j + t_{\text{start}}) \quad (34.2)$$

□

## Bibliography

- [1] R. Kress, *Linear Integral Equations*, vol. 82. in Applied Mathematical Sciences, vol. 82. New York, NY: Springer New York, 2014. doi: 10.1007/978-1-4614-9593-2.